Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Often a ground glass stopper gets stuck in the neck of a glass bottle containing $NaOH$ solution. This is due to :

(A) The presence of dirt particles in between.

(B) The formation of solid silicate in between by the reaction of SiO_2 of glass with $NaOH$.

(C) The formation of Na_2CO_3 in between by the reaction of CO_2 of air and $NaOH$.

(D) Glass contains a boron compound which forms a precipitate with the $NaOH$ solution.

Q.2

$(Si_2O_5)_n^{2n-}$ anion is obtained when:

(A) no oxygen of a SiO_4^{4-} tetrahedron is shared with another SiO_4^{4-} tetrahedron

(B) one oxygen of a SiO_4^{4-} tetrahedron is shared with another SiO_4^{4-} tetrahedron

(C) two oxygen of a SiO_4^{4-} tetrahedron are shared with another SiO_4^{4-} tetrahedron

(D) three oxygen of a SiO_4^{4-} tetrahedron are shared with another SiO_4^{4-} tetrahedron

Q.3

The gas evolved on heating CaF_2 and SiO_2 with concentrated H_2SO_4, on hydrolysis gives a white gelatinous precipitate. The precipitate is:

(A) hydrofluosilicic acid

(B) silica gel

(C) silicic acid

(D) calciumfluorosilicate

Q.4

On controlled hydrolysis and condensation, R_3SiCl yields

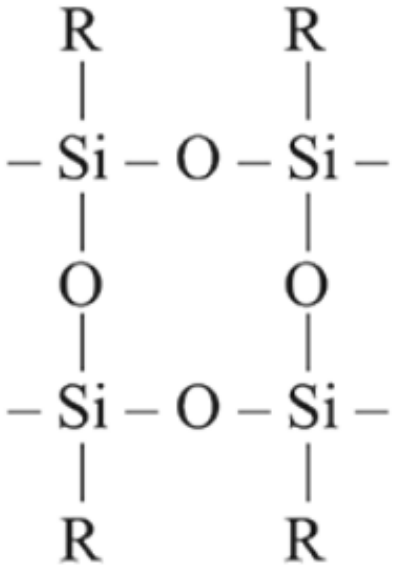
(A) $R_3Si-O-SiR_3$

(B) $(R_3Si-O-SiR_3)_n$

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(C) R_3SiOH

(D)



Q.5

When steam reacts with red hot coke to form CO_2 and hydrogen :

(A) Water acts as an oxidising agent.

(B) Water acts as a reducing agent.

(C) Carbon acts as an oxidising agent.

(D) There is no oxidation or reduction.

Q.6

Example of a three-dimensional silicate is:

(A) Zeolites

(B) Ultramarines

(C) Feldspars

(D) All of the above

Q.7

Identify the incorrect statement :

(A) In $(Si_3O_9)^{6-}$, tetrahedral SiO_4 units share two oxygen atoms.

(B) Trialkylchlorosilane on hydrolysis gives R_3SiOH

(C) $SiCl_4$ undergoes hydrolysis to give H_4SiO_4.

(D) $(Si_3O_9)^{6-}$ has cyclic structure.

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Q.8

$2RCl + Si \xrightarrow[570\ K]{Cu\ powder} R_2SiCl_2 \xrightarrow{H_2O} R_2Si(OH)_2 \xrightarrow{Polymerisation} X$, then X will be

(A) Linear long chain silicone

(B) Cyclic silicone

(C) Cross linked silicone

(D) None of these

Q.9

Among the following substituted silanes the one which will give rise to cross linked silicone polymer on hydrolysis is

(A) R_4Si

(B) R_2SiCl_2

(C) $RSiCl_3$

(D) R_3SiCl

Q.10

Bauxite ore is generally contaminated with impurity of oxides of two elements X and Y. Which of the following statement is correct?

(A) X is a non-metal and belongs to the third period while Y is a metal and belongs to the fourth period.

(B) One of two oxides has three-dimensional polymeric structure.

(C) Both (a) and (b) are correct.

(D) None of the above.

Q.11

The structure and hybridization of $Si(CH_3)_4$ is

(A) Bent, sp

(B) Trigonal, sp^2

(C) Octahedral, d^2sp^3

(D) Tetrahedral, sp^3

Q.12

Which of the following is/are not correctly matched?

(i) GeO_2 – Acidic

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(ii) PbO_2 – Amphoteric
(iii) CO - Neutral
(iv) SiO_2 – Amphoteric

(A) (i) and (iv)

(B) (iv) only

(C) (ii) only

(D) (iii) only

Q.13

Silicones contain

(A) C, O and Si

(B) C and Si

(C) O and Si

(D) C, N, O and Si

Q.14

Silanes are compounds which contain

(A) C, H and Si

(B) C and Si

(C) O and Si

(D) H and Si

Q.15

Each silicon atom in the trimer cyclic silicate has

(A) one terminal oxygen and three bridging oxygens

(B) two terminal oxygens and two bridging oxygens

(C) three terminal oxygens and one bridging oxygen

(D) all the four oxygens as the bridging oxygen

Answers & Solutions

Q.1 Answer:

The formation of solid silicate in between by the reaction of SiO_2 of glass with $NaOH$.

Solution:

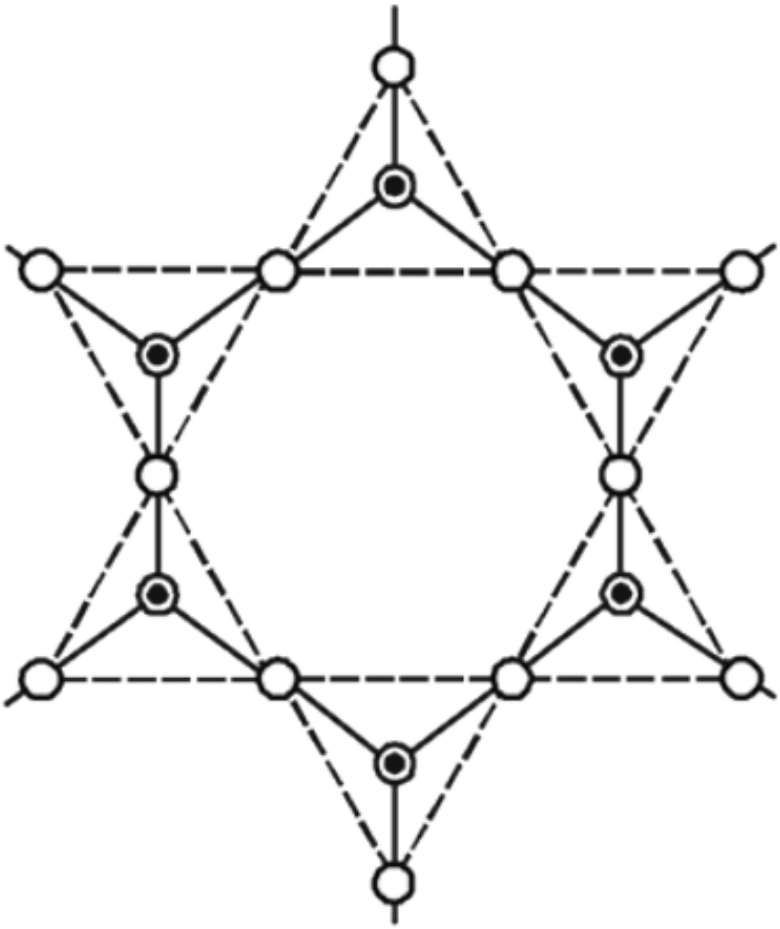
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$$SiO_2 + 2NaOH \rightarrow \underset{\text{Sodium silicate}}{Na_2SiO_3} + H_2O$$

Q.2 Answer:

three oxygen of a SiO_4^{4-} tetrahedron are shared with another SiO_4^{4-} tetrahedron

Solution:



Total No. oxygen atoms per silicon atom

$= \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + 1 = 2.5$

$\therefore$ **Formula** $Si_2O_5^{2-}$

Q.3 Answer:

calciumfluorosilicate

Solution:

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$$2CaF_2 + SiO_2 + H_2SO_4 \longrightarrow$$

$$SiF_4 + H_2O + CaSO_4 \xrightarrow{\text{hydrolysis}} CaSiF_4$$

Q.4 Answer:

$R_3Si-O-SiR_3$

Solution:

R_3SiCl **on hydrolysis forms only a dimer.** $R_3SiOH + HOSiR_3 \rightarrow R_3Si-O-SiR_3$

Q.5 Answer:

Water acts as an oxidising agent.

Solution:

$H_2O + C \longrightarrow CO_2 + H_2$

Steam

H_2O **oxidises** $C \longrightarrow CO_2$

hence H_2O **acts as oxidising agent.**

Q.6 Answer:

All of the above

Solution:

When all the four corners of a tetrahedron are shared with other tetrahedral, the three-dimensional network structure is obtained. Since all the oxygen atoms are shared, these silicates are neutral Zeolites and feldspars (Eg: albite $(NaAlSi_3O_8)$**) are three-dimensional networks of aluminosilicates.**

Ultramarines is $Na_8\left[SiAlO_4\right]_6 \cdot (S_3)_2$ **is also a 3 dimensional silicate.**

Hence, all of the are three-dimensional silicates.

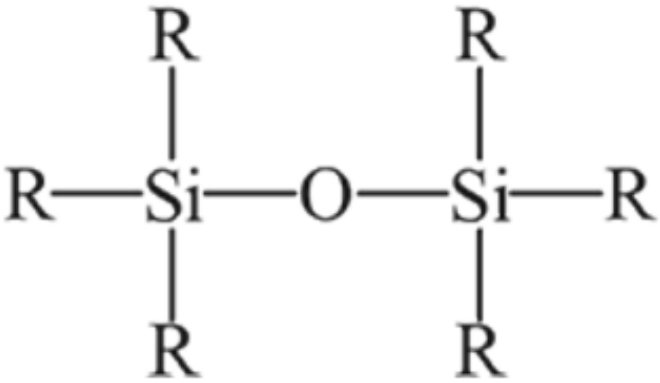
Q.7 Answer:

Trialkylchlorosilane on hydrolysis gives R_3SiOH

Solution:

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The hydrolysis of Trialkylchlorosilane R_3 SiCl yields dimer:

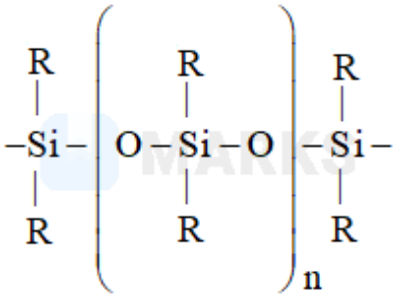


Q.8 Answer:

Linear long chain silicone

Solution:

Structure of X will be



Linear long chain polymer

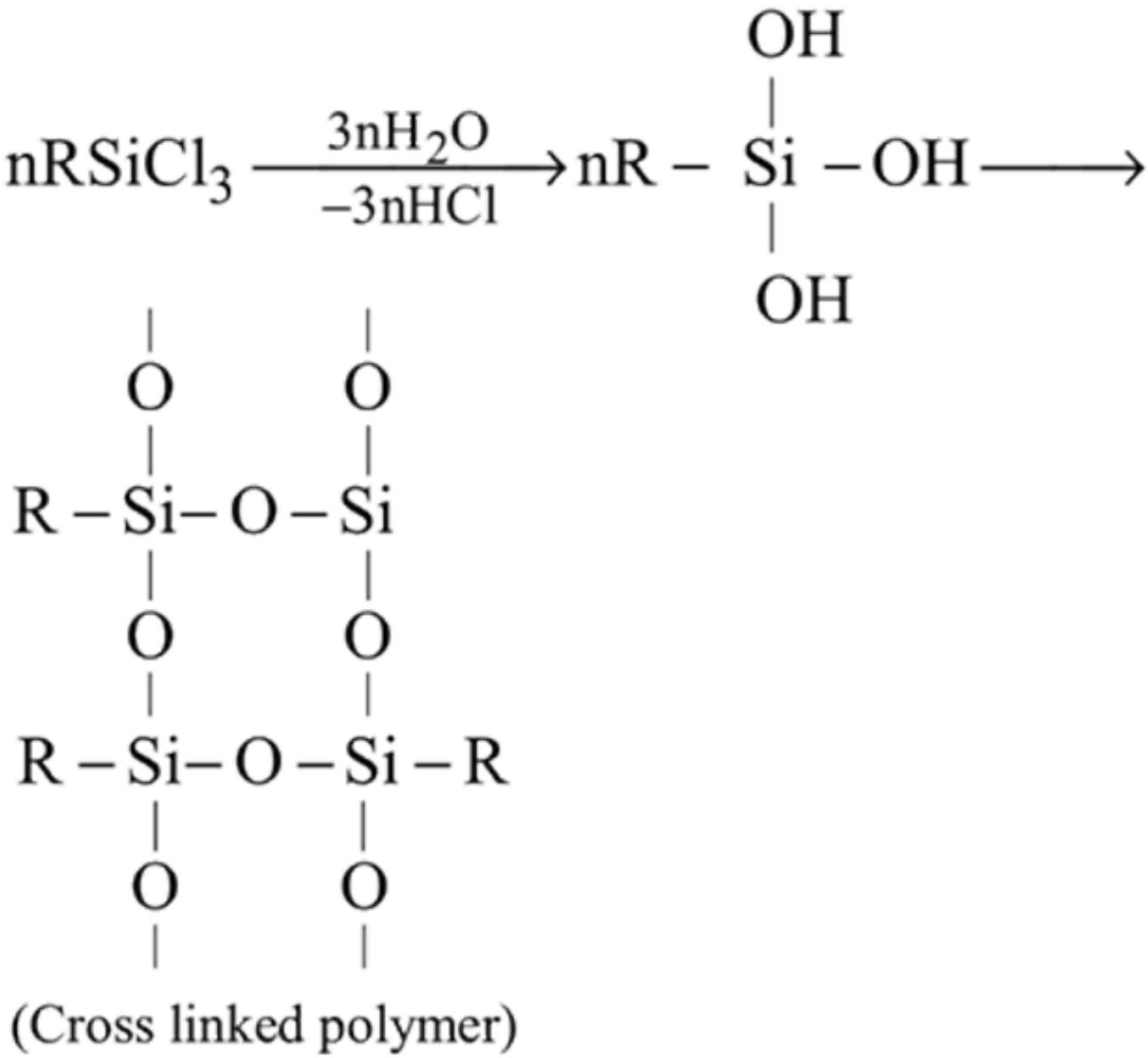
Q.9 Answer:

$RSiCl_3$

Solution:

The cross linked polymers will be formed by $RSiCl_3$

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Q.10 Answer:

Both (a) and (b) are correct.

Solution:

Two oxides present in bauxite as an impurity are SiO_2 and Fe_2O_3. Si belongs to the third period and Fe to the fourth period. SiO_2 has a three dimensional structure

Q.11 Answer:

Tetrahedral, sp^3

Solution:

Hybridisation $= \frac{1}{2}(4+4+0-0) = 4$,
sp^3 **tetrahedral.**

Q.12 Answer:

(iv) only

Solution:

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SiO_2 **is acidic oxide.**

Q.13 Answer:

C, O and Si

Solution:

C, O **and** Si

Q.14 Answer:

H and Si

Solution:

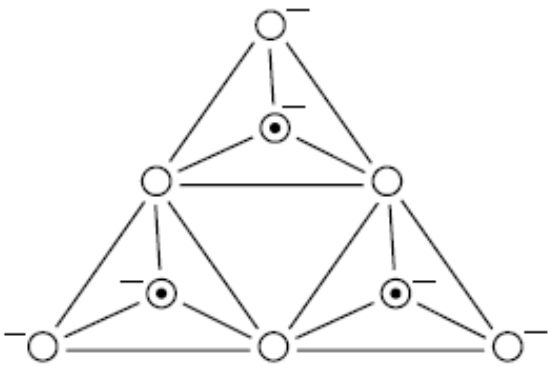
H **and** Si

Q.15 Answer:

two terminal oxygens and two bridging oxygens

Solution:

Trimeric cyclic silicate is



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